

New Project Ideas

SIF Precursor Detection - Remodel: Reports + CCTV Surveillance + AI Risk

GOARC Safety 1.0 Prototype - Oil India Limited | PS 26165

Ideas at a Glance - What Changes

In one sentence: Keep the report flow you already have, add a CCTV page for live footage, let AI do all risk math, lock every change in an un-deletable log, add auto-reports and Site Risk Index, and put an OIL landing page first.

- **Reports stay - but simpler:** User only writes *what happened, where, under whose supervision*. No manual risk picker. AI generates risk, status, category, LSR and shows draft.
- **CCTV Surveillance Page (new):** Upload a CCTV clip or select sample footage. AI watches for PPE, zone entry, fall, proximity, fire. Gives verdict: Warning / Serious / Critical.
- **AI owns the risk:** No dropdown for High/Medium/Low. AI decides from narrative or video.
- **Immutable audit log (new):** Every addition / edit / deletion / status change is appended to report_logs with timestamp, actor, old vs new - never overwritten.
- **Auto-alert on serious injury:** If AI verdict is Critical, backend instantly alerts safety personnel on web + phone app + optional speaker mock.
- **Two report types, one DB:** Text reports and CCTV incidents share same tables with source: MANUAL vs CCTV.
- **Officer gets notification:** Push via FCM/APNs to officer on duty with snapshot + 30s clip.

UI Remodel - First Page is Now an OIL Landing Page (New)

Because the platform is now **CCTV Surveillance + Report SIF**, the old Dashboard-first UI is replaced. **First page (/) is now a normal informational landing page about the project and how it works for Oil India Company** - like a product site, not a dashboard. Dashboard moves to **/dashboard**.

Landing content: Hero (SIF Precursor Detection, Industrial Safety Dashboard, 600 reports across 15 sites, GOARC amber CTA), Problem (why OIL needs SIF triage), How it works (3 steps: Report or CCTV -> AI Vision -> Risk + Alert), For OIL (drilling / pipeline / refinery use cases), Key features (auto-report 9 fields, Site Risk Index heatmap, immutable log, phone push), CTA (View Dashboard, Watch CCTV demo). All in GOARC dark/light alternating bands with wave dividers (85px).

Flow: Visitor lands on / -> reads about OIL SIF for Oil India -> clicks *Enter Platform* -> goes to /dashboard or /cctv. Navbar stays (OIL badge) but landing has its own hero. Keeps Dashboard, Reports, CCTV, AI Prediction as app pages behind the landing.

Three New Intelligent Features (Added)

1 and 2. Auto-Generated Incident Report from CCTV - When CCTV detects a hazard, the system not only alerts the officer but automatically creates a detailed incident report with 10 fields:

- **What happened** - description (e.g., Worker without helmet entered Zone B, then fell).
- **Where it happened** - factory zone, location, or camera ID (Zone B, Drill Deck, Cam 2).
- **When it happened** - date and timestamp (30 Aug 2026, 14:32:12).
- **Why it happened** - possible cause (missing PPE + unauthorized entry + no isolation).
- **How it happened** - sequence (enter -> PPE violation -> proximity -> fall -> dwell).
- **Severity/Risk level** - Warning / Serious / Critical.

- **People/objects involved** - Worker 23, helmet No, vest Yes, machinery.
- **Corrective measures** - immediate actions (isolate area, evacuate, provide PPE).
- **Preventive measures** - recommendations to prevent recurrence (retrain, add barrier, update PTW).
- **CCTV evidence** - snapshot + 30s before / 60s after clip.

The generated report is **reviewed by the responsible officer** (Approve / Edit) and then stored alongside the incident and its full **immutable audit history**.

3. Site Risk Index - Prioritize by History

Analyzes historical incident data using three weighted factors and visualizes high-risk zones on a site map/heatmap so HSE teams focus preventive action where it matters most.

- **Frequency**: Count per zone in window (90 days).
- **Severity potential**: Weighted: Critical=10, Serious=5, Warning=1.
- **Recency**: Exponential decay: last 7d x1.0, 30d x0.6, 90d x0.3.
- **Formula (demo)**: $RiskIndex = \sum(severity_weight * recency_decay) * \log(1+frequency) * 10$ normalized 0-100. Visualized as heatmap (green->amber->red) + ranked table.
- **Action**: HSE sees Top 3 risky zones this month + trend line; clicking a zone filters reports.

Remodeled Report Flow - Simple for User, Strict for System

1. **User inputs (3 fields)**: Description, Where, Supervised by (no Risk picker, 5000-char guard).
2. **AI drafts**: POST to 8001 -> draft category/risk/status/LSR/probability/actions -> GOARC draft card with Submit or Edit.
3. **User action**: Submit saves to reports + classifications + report_logs (immutable); Edit re-runs AI.
4. **CCTV auto-draft**: Video -> Vision -> same 9-field report auto-created + push + speaker if Critical.

New Page - CCTV Surveillance

Page /cctv: camera selector + Upload clip, video with AI boxes (green/red PPE), zone overlay, live verdict (Warning/Serious/Critical), incident table + clip player. Upload MP4 -> Vision (YOLOv8 + tracking + pose) -> severity (2-sec dwell) -> CCTV report + push + speaker mock.

Immutable Log Report - Tamper-Proof Diary

Table report_logs (id PK, report_id FK, action CREATE/UPDATE/DELETE/STATUS_CHANGE/CCTV_CREATE, actor, timestamp, old_value JSON, new_value JSON, source MANUAL/CCTV) append-only, no DELETE API, UI History timeline read-only, only code migration can alter.

What Each Group Must Do

App Development (new)

Easy: Officer phone app (login, live feed, push). Technical: Expo + FCM/APNs, push_token in users, 4 screens, JWT, 15s poll fallback.

Key files: mobile/ repo, feat/mobile-push

Frontend

Easy: Simplify Add Report + new /cctv + new landing + auto-report preview + Risk Index heatmap. Technical: React Router /cctv (canvas bbox, zone editor, verdict pill), Report History, /api/cctv/events live poll, new Landing.jsx (hero + 5 sections + wave), Site Risk heatmap.

Key files: pages/Cctv.jsx, Landing.jsx, RiskIndex.jsx, api.js

Backend

Easy: AI is decider + log everything + auto-report. Technical: report_logs model, remove manual risk, new endpoints POST /api/cctv/events + /incidents/notify + /acknowledge, FCM fan-out, 60s escalation, auto-generate 9-field report on CCTV.

Key files: main.py, crud.py, models.py, schemas.py

Database

Easy: Keep reports as source, add diary + Site Index. Technical: report_logs + source enum + push_token + subscriptions, site_risk_scores (zone, index, frequency, severity_sum, recency), indexes, seed --force.

Key files: init.sql, models.py, seed.py, migrations

AI

Easy: See hazards + write report + score risk. Technical: YOLOv8 + ByteTrack + pose for auto-report 9 fields + Site Risk Index calc (frequency x severity x recency).

Key files: ai/vision.py, analyser.py, 8001/8002

Docker / DevOps

Easy: Run video AI and push. Technical: ai-vision service (libgomp1 + ultralytics), env CCTV_RTSP_URL + FCM_SERVER_KEY, 8002, healthchecks, start.sh warm-up.

Key files: docker-compose.yml, Dockerfiles, start.sh

Git

Easy: Keep branches tidy. Technical: feat/report-remodel, feat/cctv-auto-report, feat/site-risk-index, feat/landing-page, feat/mobile-push.

Key files: Branches, PRs, tag v0.5

QA / Demo

Easy: Prove it to judges. Technical: 2 canned clips -> auto 9-field report push -> ACK -> History + Site Risk heatmap shows Zone B red + landing demo.

Key files: data/cctv/, demo script

Stack: React 19 + Vite + GOARC + FastAPI + SQLAlchemy + YOLOv8 + sklearn + PostgreSQL/SQLite + Docker + Expo - Ports 5173/8000/8001/8002/5432